

AI-Driven PM_{2.5} Prediction Using Satellite and Meteorological Data

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Abstract—(AI) (ML) and deep learning (DL) models based on weather observations and satellite-derived Aerosol Optical Depth (AOD). Satellite observed Aerosol Optical Depth (AOD), together with meteorological conditions are used to derive PM_{2.5} concentration near surface. The datasets used in this research were collected from monitoring stations in the District of Columbia, Maryland, and Virginia (DMV) region between 2019 and 2022 and synchronized into 60-minute and 30-minute intervals. Several ML models were tested alongside five DL architectures: Simple RNN, LSTM, GRU, Bi-LSTM, and Bi-GRU. The models were evaluated using R^2 score, RMSE, and MAE to assess predictive accuracy. Notably, the Lasso regression model achieved a peak performance of $R^2 = 0.963$ and an RMSE of 0.946 at the Beltsville station. Among deep learning architectures, Bi-GRU had the best performance on the 60-minute dataset, while LSTM gave the highest predictive performance on the 30-minute dataset. These findings can help with more accurate air quality monitoring and better environmental and public health decisions.

Index Terms—PM_{2.5} prediction, Air Quality Monitoring, Aerosol Optical Depth, Machine Learning, Deep Learning, Time Series Forecasting, Satellite Data

I. INTRODUCTION

Air pollution caused by fine particulate matter of diameter less than 2.5 micrometers ($PM_{2.5}$) keeps creating serious risks for human health and the environment [1], [2]. $PM_{2.5}$ grains are small enough to slip deep into the lungs and bloodstream, which potentially causes respiratory diseases, cardiovascular problems, and other health complications [3], [4]. Oftentimes, prolonged exposure to $PM_{2.5}$ increases the severity of these diseases.

Although continuous monitoring of $PM_{2.5}$ is vital for public health and environmental policy, many regions, especially developing countries and rural areas, still lack enough ground-based monitoring stations due to high installation, operational and maintenance costs [5], [6]. Satellite-derived AOD is therefore widely used to

estimate surface-level $PM_{2.5}$, and many Artificial Intelligence (AI) models have been proposed for this task. However, accurate prediction remains difficult because $PM_{2.5}$ depends not only on AOD but also on complex nonlinear interactions with meteorological variables such as temperature, relative humidity, wind speed, and wind direction [7]–[10]. These spatiotemporal relationships limit conventional AI models and motivate more advanced approaches for reliable $PM_{2.5}$ retrieval and prediction.

This study aims to develop an AI model that can accurately estimate surface-level $PM_{2.5}$ in regions with limited or no monitoring stations while effectively capturing the complex nonlinear relationships between $PM_{2.5}$, AOD, and meteorological variables. To achieve this, multiple ML models, including DL models are systematically trained and evaluated through several development phases using datasets with different temporal resolutions collected under diverse environmental conditions. The comparative evaluation of these models across different temporal resolutions and environmental settings will identify the most accurate and robust model for estimating $PM_{2.5}$ in locations where ground-based monitoring data are unavailable.

To address the challenges of sparse monitoring and complex AOD–meteorology– $PM_{2.5}$ relationships, a data-driven AI framework was developed using NASA GEO-LEO Dark Target AOD product [11] integrated with Environmental Protection Agency (EPA) ground observations (temperature, relative humidity, wind speed) [12]. Data from Howard, Padonia, and Beltsville in Maryland were used to capture different environmental conditions and improve the models’ ability to learn atmospheric variability. Since satellite and ground observations are recorded at different temporal resolutions, datasets were matched and interpolated to 60-min and 30-min intervals to assess temporal-resolution effects. Models were trained in three progressive phases, baseline learning, feature engineering, and

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model optimization, to iteratively capture temporal and environmental patterns; the best models were identified using standard statistical metrics for reliable $PM_{2.5}$ estimation where ground monitors are absent.

The novelty is twofold: a multi-phase training strategy combined with evaluation at multiple temporal resolutions to progressively capture complex AOD–meteorology– $PM_{2.5}$ relationships, and application to three Maryland sites to test transferability across diverse local atmospheres. This approach yields improved learning of temporal patterns and more reliable $PM_{2.5}$ estimates for regions with sparse ground monitoring.

The remainder of this paper is organized as follows. Section II reviews related work, Section III presents the proposed methodology, Section IV reports the experimental results, Section V discusses the findings, Section VI outlines the limitations, Section VII suggests future work, and Section VIII concludes the paper.

II. RELATED WORKS

Although all the reviewed articles discuss the application of ML for the estimation or prediction of $PM_{2.5}$, each paper has different objectives, models, datasets, and methodology. The discussed articles present various ways of implementing ensemble learning, DL, hybrid, and tree algorithms to improve the accuracy of air quality models and better understand environmental processes. Danesh Yazdi et al. proposed an ensemble machine learning framework to estimate daily $PM_{2.5}$ concentrations using satellite-derived AOD, land-use, and meteorological data. Their model combined Random Forest (RF), Gradient Boosting Machine (GBM), Deep Neural Network (DNN), and K-Nearest Neighbor (KNN) with a Generalized Additive Model (GAM), achieving a 10-fold cross-validated Coefficient of determination (R^2) of 0.828. The ensemble improved prediction accuracy by integrating multiple learning algorithms, but its spatial prediction performance was lower and required numerous environmental variables, increasing computational complexity [1].

Zamani Joharestani et al. compared Random Forest, XGBoost, and Deep Neural Networks for predicting $PM_{2.5}$ concentrations using remote sensing, meteorological, and geographical data. XGBoost achieved the best performance with an R^2 of 0.81 and improved model interpretability through feature selection. However, the effectiveness of AOD was limited because of the large number of missing observations [2].

Nath et al. proposed a four-stage hybrid framework that combines statistical, ML, DL, and satellite-derived AOD data to estimate $PM_{2.5}$ concentrations in areas with limited monitoring stations. Random Forest Regression provided the best spatial interpolation, while

the optimal temporal forecasting model differed across cities. Although the framework effectively addressed data sparsity, its multi-stage design increased computational complexity [13].

Adong et al. evaluated Multivariate Linear Regression, RF, and Multilayer Perceptron models using AOD together with low-cost and reference-grade monitoring stations. RF produced the best prediction accuracy and demonstrated that low-cost sensors can support satellite-based $PM_{2.5}$ estimation in regions with limited monitoring infrastructure. However, missing satellite observations and limited monitoring stations reduced model robustness [14].

Yu et al. introduced a three-stage Deep Ensemble Machine Learning (DEML) framework that combines GBM, RF, Support Vector Machine, XGBoost, and meta-learning techniques for $PM_{2.5}$ estimation. The framework outperformed individual machine learning models by improving prediction accuracy and generalization. However, its three-stage architecture requires greater computational resources and longer training times [15].

Li et al. developed a spatiotemporally weighted tree-based algorithm that improves both the accuracy and interpretability of $PM_{2.5}$ prediction by accounting for geographical and temporal variations. The model provides better insight into feature importance than conventional tree-based methods. However, the additional spatial and temporal weighting increases computational complexity [16].

Aditya et al. developed a lightweight air pollution monitoring system that combines Logistic Regression for pollution classification with Autoregression for forecasting future $PM_{2.5}$ levels. The approach is computationally efficient and suitable for real-time applications, but its prediction accuracy is lower than that of modern ensemble and deep learning methods [17].

Bochenek and Ustrnul reviewed over 500 studies on machine learning applications in weather prediction and climate analysis. They identified Random Forest, Artificial Neural Networks, DL, SVM, and XGBoost as the most widely used techniques for environmental forecasting. Although the review summarizes current machine learning approaches and future research directions, it does not experimentally compare $PM_{2.5}$ prediction models [18].

Together, the reviewed articles indicate that ensemble learning, DL, and hybrid ML methods could help improve predictions of $PM_{2.5}$ by considering the nonlinear relationships between $PM_{2.5}$ and meteorological variables, satellite AOD, and historical air quality data [1], [2], [13], [15], [16], shown in Table I. However, the studies note a number of challenges and limitations, such as the lack of satellite AOD information [2], [14], the computational complexity of ensemble and hybrid

TABLE I: Summary Comparison of Related Works

Ref	Approach	Results	Novelty	Limitations	Target Scope
[1]	Ensemble ML (RF, GBM, DNN, KNN + GAM)	10-fold CV $R^2 = 0.828$	Combines ensemble learning with satellite AOD, land-use, and meteorological data	High computational complexity; lower spatial prediction accuracy	Greater London
[2]	RF, XGBoost, and DNN using multisource remote sensing	XGBoost achieved $R^2 = 0.81$	Feature importance analysis using remote sensing and meteorological data	Large amount of missing AOD observations	Tehran, Iran
[3]	Hybrid statistical, ML, DL, and AOD framework	RF best for spatial interpolation; city-dependent temporal models	Four-stage framework for sparse monitoring regions	High computational complexity	Kolkata, Bengaluru, Mumbai
[4]	MLR, RF, and MLP with satellite AOD and low-cost sensors	RF achieved the highest prediction accuracy	Combines low-cost monitoring stations with satellite observations	Cloud cover and limited monitoring stations	Kampala, Uganda
[5]	Three-stage Deep Ensemble ML (GBM, RF, SVM, XGBoost)	Improved prediction accuracy and model generalization	Stacked ensemble learning architecture	Long training time and high computational cost	Italy
[6]	Spatiotemporally Weighted Tree-Based Algorithm	Improved prediction accuracy and interpretability	Models spatial and temporal heterogeneity simultaneously	Computationally intensive	China
[7]	Logistic Regression with Autoregression	Efficient real-time pollution prediction	Lightweight forecasting framework	Lower prediction accuracy than ensemble and deep learning methods	General air pollution monitoring
[8]	Review of ML methods for weather and climate prediction	Surveyed over 500 machine learning studies	Comprehensive review of environmental forecasting techniques	No experimental comparison of $PM_{2.5}$ prediction models	Weather and climate forecasting

approaches [13], [15], [16], the lack of air pollution monitoring resources in specific areas [13], [14], and the challenge of capturing generalizability across different locations and environments [1], [16]. Thus, there is a need to develop computationally efficient and transferable ML models that could predict $PM_{2.5}$ levels based on various environmental data.

III. METHODOLOGY

As illustrated in Fig. 1, the process is divided into five primary stages: data integration, temporal synchronization, data cleaning and pre-processing, a three-phase model training, and final performance evaluation.

A. Data Description and Integration

The data used in this study were collected from monitoring stations in the DMV region between 2019 and 2022. Data from three monitoring locations: Howard, Padonia, and Beltsville were used to develop the prediction models. Two complementary data sources were integrated to construct the final dataset.

The EPA [12] data source provides hourly ground-based meteorological observations, including $PM_{2.5}$ concentration (the target variable), temperature, relative humidity, and wind speed. The GEO-LEO Dark Target [11] dataset consists of AOD measurements collected at 10-minute intervals. Given the differences in temporal resolution, these two sources were synchronized before being merged.

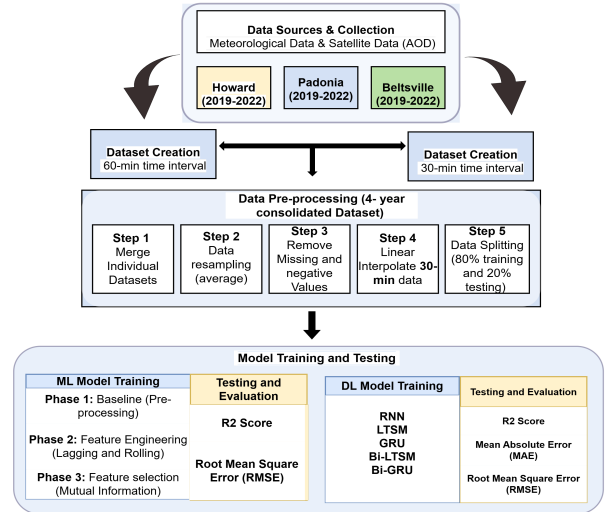


Fig. 1: Process Flow Diagram of this work

B. Temporal Synchronization

To ensure consistency, both data sources were synchronized to common temporal resolutions prior to integration. Two datasets were generated for model development: a 60-minute (hourly) dataset and a 30-minute dataset.

For the 60-minute dataset, the meteorological observations were maintained at an hourly resolution, while the original 10-minute AOD measurements were aggregated to hourly intervals. This ensured that observations from

both datasets corresponded to the same timestamps.

To generate the 30-minute dataset, the synchronized hourly observations from both datasets were used as the baseline. Linear interpolation [19] was then applied independently to the meteorological variables and the AOD measurements to estimate values at intermediate 30-minute timestamps. This process produced a higher temporal resolution dataset while preserving temporal alignment between all variables.

C. Data Merging

Following temporal synchronization, the meteorological and satellite-derived AOD datasets were merged based on their matching timestamps. Observations sharing identical timestamps were combined into a single record, ensuring that all predictor variables corresponded to the same observation time.

Each merged observation contained the following variables: $PM_{2.5}$ concentration, Temperature, Relative humidity, Wind speed and AOD.

D. Data Cleaning and Pre-processing

Before model development, the merged dataset underwent several preprocessing steps. The dataset was inspected for missing values, and negative and invalid observations were identified and removed. **final data count**

TABLE II: Summary of Data Records After Resampling, Merging, and Cleaning

Stage	Howard	Padonia	Beltsville
Initial AOD	43,731	47,477	8,450
Matched Records	6,995	7,406	8,164
Negative/Null AOD Removed	175	340	0
Final Cleaned Dataset	6,820	7,066	8,164

Finally, the dataset was divided into training and testing sets, with 80% allocated for model training and the remaining 20% reserved for model testing.

E. Model Training

Following preprocessing, a total of 14 ML and DL models including linear, tree-based, ensemble, and recurrent neural network architectures, were trained and evaluated on both datasets. ML models followed a three-phase framework: baseline modeling (phase 1), feature engineering (phase 2), and feature selection (phase 3), while DL models were trained directly on the preprocessed datasets.

1) *ML Models*: Nine supervised ML regression models representing three algorithm categories were evaluated for $PM_{2.5}$ prediction: linear models (Linear Regression, Ridge, Lasso), tree-based models (Random Forest, Extra Trees, and Gradient Boosting), and ensemble models (CatBoost, XGBoost, and LightGBM).

Linear models estimate the relationship between the input features and target variable by fitting a linear function, while Ridge and Lasso apply regularization to reduce overfitting and improve generalization by shrinking regression coefficients. Tree-based models recursively partition the feature space into homogeneous regions, enabling them to capture complex nonlinear relationships while remaining robust to outliers and noisy data. Ensemble learning models iteratively combine multiple weak learners, where each successive model focuses on correcting the prediction errors of previous learners, resulting in improved prediction accuracy and better representation of nonlinear patterns.

To systematically evaluate the contribution of temporal information and feature relevance, all ML models were trained using a three-phase framework.

Phase 1 (Baseline Modeling): All models were initially trained using the original features, including AOD, temperature, relative humidity, and wind speed, establishing a baseline for comparison.

Phase 2 (Feature Engineering): Temporal dependencies were incorporated by generating lag and rolling statistical features. Lagging is a time-series data technique that allows models to use past values to learn patterns. This leads to predictions based on trends of the past. Rolling features summarize statistics such as the mean and standard deviation over a given window to capture momentum and volatility. Lag features of 1, 2, 3, 6, 12, 24, and 48 hours enabled the models to learn from historical observations, while rolling statistics (mean and standard deviation) computed over 6-, 12-, 24-, and 72-hour windows captured short- and long-term temporal trends and variability. This phase expanded the feature space to 80 features.

Phase 3 (Feature Selection): Mutual Information (MI) was employed to quantify the dependency between each feature and the target variable by measuring the reduction in prediction uncertainty [20]. The top 40 features with the highest MI scores were selected and used to retrain all ML models, reducing feature redundancy while retaining the most informative predictors.

2) *DL models*: For the DL portion of the study, since models such as SimpleRNN, LSTM, and GRU are designed to learn from sequences, the original data were arranged into input-output pairs using a sliding window method. The models were also tested at two different time intervals, 1-hour intervals and 30-minute intervals. This was done to see whether using more frequent time steps would improve the model's ability to predict $PM_{2.5}$. A sliding window of 10 time steps was applied to the dataset. This means that the model used 10 previous time steps of data to predict the $PM_{2.5}$ concentration at the next time step. Each DL model was trained for 50 epochs. After each epoch, the model was

evaluated on both the training and testing data. Mean Absolute Error (MAE) and R^2 scores were recorded after every epoch so that changes in model performance could be tracked throughout training. This made it possible to see whether the model was improving, overfitting, or beginning to level off. Five deep learning models were tested: SimpleRNN, LSTM, GRU, Bidirectional LSTM (Bi-LSTM), and Bidirectional GRU (Bi-GRU). Each model was tested using an uni-variate setup, where the model used only previous $PM_{2.5}$ values to predict future $PM_{2.5}$ concentration.

F. Model Testing and Evaluation

Following training, the ML models were evaluated using the coefficient of determination (R^2) and Root Mean Square Error (RMSE), whereas the DL models were evaluated using R^2 , Mean Squared Error (MSE), and Mean Absolute Error (MAE). MSE was used as the DL loss function and later converted to RMSE. The evaluation metrics are defined in (1)–(4).

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (1)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (2)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (3)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (4)$$

where y_i , $\hat{y}_i$, $\bar{y}$, and n denote the observed value, predicted value, mean observed value, and total number of samples, respectively.

IV. RESULTS

A. ML Models Results

In Table III, across almost all models and locations, R^2 is seen to have improved the most from Phase 1 to Phase 2. This shows that lagging and rolling features had the largest impact on model performance. By adding information about recent $PM_{2.5}$ patterns, the models were able to make more accurate predictions. The change from Phase 2 to Phase 3 was smaller and less consistent. In some cases, feature selection significantly improved performance, while in other cases the improvement was minor. This suggests that reducing the number of features did not always lead to a major increase in R^2 . Beltsville had the strongest overall model performance. The highest R^2 score was 0.963, achieved by Lasso Regression in Phase 3 using the 30-minute dataset. At Howard, CatBoost produced the strongest result, with an R^2

score of 0.885 under the same 30-minute Phase 3 setup. Overall, the 30-minute dataset usually performed better than the 60-minute dataset.

Figure 2 shows actual-value vs. predicted-value plots, comparing the true $PM_{2.5}$ measurements against each model's predictions across the three stations (Padonia, Beltsville, Howard) at both 60-minute and 30-minute intervals. Within each station's block, the left column shows model predictions from Phase 1 (the baseline models), while the right column shows predictions from Phase 3, after feature engineering and feature selection. Each plot also includes a line of best fit. The closer this line tracks to a 45-degree diagonal, the more accurate the model's predictions are, since points falling exactly on that line would mean predicted values perfectly match actual values. Moving from Phase 1 to Phase 3, there's a clear shift in the best-fit lines toward a steeper, more 45-degree-aligned slope, which lines up with the higher R^2 values reported in each subplot, confirming that feature engineering improved model accuracy.

Figure 3 compares RMSE across three models—Random Forest Regressor, Linear Regression, and CatBoost—for each station, shown at both hourly and 30-minute intervals. For each station, "Baseline" bars represent RMSE before feature engineering, while "Final" bars represent RMSE after feature engineering and selection. Since lower RMSE indicates lower margin of errors, the consistent drop from Baseline to Final bars across nearly all stations and models demonstrates that feature engineering meaningfully reduced prediction error. This trend holds at both time intervals, reinforcing that the improvements seen in Figure 2 tighter best-fit lines are backed by a measurable reduction in error, not just a visual one.

Table IV shows the RMSE results for the same machine learning models, locations, time intervals, and three-phase process discussed in Table III. Overall, the RMSE and R^2 results support the same pattern. Accuracy increased the most from Phase 1 to Phase 2 across most models and locations. This confirms that the lagged and rolling-window features had the most meaningful impact on models' performance. Several models produced the lowest RMSE values depending on the site and time interval. At the 60-minute interval, CatBoost had the lowest RMSE at Howard and Beltsville under Phase 3, with values of 1.970 and 1.50. At the 30-minute interval, Random Forest had the lowest RMSE at Beltsville under the baseline phase, with an RMSE of 0.8434. Extra Trees had the lowest RMSE at Padonia under Phase 3, with an RMSE of 1.206.

B. DL Models Results

Table V shows that across all DL models, the results were very close within each site and time interval. For

TABLE III: Result Analysis [R-squared (R²)] for 60-minute and 30-minute datasets.

ML Models	60 Minutes Dataset									30 Minutes Dataset								
	Baseline (Phase 1)			Feature Engineering (Phase 2)			Feature Selection (Phase 3)			Baseline (Phase 1)			Feature Engineering (Phase 2)			Feature Selection (Phase 3)		
	Howard	Padonia	Beltsville	Howard	Padonia	Beltsville	Howard	Padonia	Beltsville	Howard	Padonia	Beltsville	Howard	Padonia	Beltsville	Howard	Padonia	Beltsville
Random Forest	0.459	0.486	0.546	0.722	0.633	0.653	0.725	0.635	0.690	0.512	0.761	0.801	0.855	0.893	0.882	0.864	0.900	0.917
Extra Trees	0.500	0.540	0.637	0.715	0.633	0.673	0.7245	0.640	0.710	0.510	0.844	0.893	0.870	0.895	0.906	0.879	0.905	0.917
Gradient Boosting	0.430	0.443	0.405	0.703	0.630	0.677	0.7171	0.629	0.700	0.469	0.510	0.441	0.837	0.843	0.812	0.843	0.845	0.877
Linear Regression	0.285	0.206	0.168	0.722	0.647	0.717	0.7237	0.651	0.718	0.338	0.249	0.190	0.853	0.856	0.890	0.854	0.862	0.891
Ridge Regression	0.285	0.206	0.168	0.722	0.647	0.717	0.7237	0.651	0.718	0.338	0.249	0.190	0.854	0.865	0.890	0.854	0.862	0.891
Lasso Regression	0.198	0.206	0.168	0.634	0.651	0.719	0.724	0.650	0.726	0.217	0.249	0.190	0.799	0.864	0.963	0.854	0.859	0.896
XGBoost	0.4688	0.481	0.389	0.717	0.634	0.637	0.720	0.639	0.701	0.463	0.483	0.442	0.863	0.841	0.848	0.848	0.846	0.876
LightGBM	0.480	0.489	0.528	0.747	0.658	0.693	0.749	0.655	0.739	0.479	0.481	0.619	0.876	0.884	0.944	0.872	0.891	0.915
CatBoost	0.495	0.406	0.564	0.756	0.656	0.687	0.754	0.665	0.718	0.498	0.650	0.677	0.885	0.892	0.841	0.881	0.897	0.914

TABLE IV: Error Analysis [RMSE] for 60-minute and 30-minute datasets.

ML Models	60 Minutes Dataset									30 Minutes Dataset								
	Baseline (Phase 1)			Feature Engineering (Phase 2)			Feature Selection (Phase 3)			Baseline (Phase 1)			Feature Engineering (Phase 2)			Feature Selection (Phase 3)		
	Howard	Padonia	Beltsville	Howard	Padonia	Beltsville	Howard	Padonia	Beltsville	Howard	Padonia	Beltsville	Howard	Padonia	Beltsville	Howard	Padonia	Beltsville
Random Forest	3.260	3.487	2.681	2.102	2.460	1.753	2.0932	2.451	1.658	3.015	2.264	1.737	1.470	1.279	1.008	1.423	1.238	0.8434
Extra Trees	3.134	3.244	2.423	2.129	2.460	1.702	2.0954	2.436	1.604	3.020	1.831	1.273	1.393	1.268	0.900	1.342	1.206	0.8424
Gradient Boosting	3.346	3.557	3.031	2.172	2.470	1.691	2.1234	2.472	1.639	3.145	3.346	2.939	1.558	1.548	1.272	1.530	1.537	1.029
Linear Regression	3.748	4.264	3.616	2.104	2.412	1.583	2.0982	2.399	1.580	3.512	4.021	3.504	1.478	1.438	0.973	1.476	1.453	0.9673
Ridge Regression	3.748	4.264	3.616	2.103	2.411	1.583	2.0983	2.398	1.580	3.512	4.021	3.504	1.478	1.438	0.973	1.476	1.453	0.9668
Lasso Regression	3.968	4.263	3.616	2.412	2.398	1.578	2.0967	2.406	1.573	3.821	4.020	3.504	1.733	1.442	0.963	1.476	1.470	0.9460
XGBoost	3.231	3.446	3.074	2.120	2.467	1.792	2.109	2.438	1.627	3.163	3.335	2.908	1.429	1.555	1.143	1.504	1.533	1.038
LightGBM	3.197	3.421	2.724	2.007	2.374	1.649	1.998	2.383	1.521	3.110	3.446	2.404	1.358	1.329	0.944	1.383	1.291	0.8715
CatBoost	3.150	4.380	2.614	1.970	2.381	1.665	1.979	2.348	1.50	3.059	2.746	2.212	1.311	1.285	1.167	1.334	1.257	0.8622

TABLE V: DL Result Analysis: 60-Min and 30-Min

DL Models	60 Minute Dataset						30 Minute Dataset					
	R^2		RMSE		MAE		R^2		RMSE		MAE	
	Howard	Padonia	Beltsville	Howard	Padonia	Beltsville	Howard	Padonia	Beltsville	Howard	Padonia	Beltsville
SimpleRNN	0.528	0.344	0.601	2.736	3.152	1.885	0.836	0.752	0.872	1.575	1.867	1.045
LSTM	0.538	0.444	0.605	2.702	3.012	1.876	0.862	0.833	0.881	1.445	1.596	1.010
GRU	0.532	0.448	0.606	2.725	3.004	1.875	0.849	0.826	0.877	1.513	1.627	1.024
Bi-LSTM	0.531	0.417	0.597	2.727	3.083	1.892	0.847	0.810	0.876	1.524	1.703	1.030
Bi-GRU	0.544	0.462	0.603	2.689	2.962	1.881	0.859	0.831	0.884	1.460	1.604	0.9977

example, at the 60-minute interval, the R^2 scores ranged from 0.528 to 0.544 at Howard and from 0.597 to 0.606 at Beltsville. This shows that the choice of deep learning architecture did not make a major difference when the models were only using past $PM_{2.5}$ values. Bi-GRU and LSTM usually performed slightly better than the other models, with somewhat higher R^2 scores and lower error values. However, the differences between the models were small. The biggest changes came from when the data changed from 60-minute intervals to 30-minute intervals, where all five models improved noticeably. For example, SimpleRNN at Howard increased from an R^2 score of 0.528 to 0.836.

Figure 4 shows the results of the two Bi-GRU models for Beltsville. At the 30-minute interval, the testing MAE settles around 0.5–0.6, while the testing R^2 reaches approximately 0.86–0.89. The training and testing curves also remain fairly close, showing that the model performs well on unseen data. In comparison, the 1-hour model has a much higher testing MAE of about 1.25–1.35, and its testing R^2 levels off near 0.60. Although both models show some difference between training and testing performance, the gap is more noticeable with the hourly data. The training performance continues to improve, while the testing results plateau, suggesting that the model is learning patterns that do not generalize as well. Overall, the results show that the finer 30-minute interval helps the Bi-GRU better capture changes in $PM_{2.5}$ at the

Beltsville site. Averaging the data into hourly intervals may smooth out important short-term patterns that the model needs to make accurate predictions.

V. DISCUSSION

Table III shows that the largest improvement in R^2 occurred between Phases 1 and 2, while the gains from Phase 2 to Phase 3 were generally smaller. This pattern directly addresses the nonlinear-relationship component of the study's aim by suggesting that recent $PM_{2.5}$ levels were among the strongest predictors of future concentrations, and that capturing temporal dependencies mattered more than reducing the feature set. Lag features gave the models access to previous $PM_{2.5}$ values, while rolling means and standard deviations captured short-term trends and variability. These features helped represent the gradual buildup and dispersion of pollutants, which could not be fully captured using only current meteorological conditions and AOD alone. This reinforces the premise that $PM_{2.5}$ estimation from AOD and weather data requires modeling beyond simple linear associations. Feature selection in Phase 3 mainly removed redundant or less useful features, which explains why its impact varied across models and sites.

The 30-minute dataset generally outperformed the 60-minute dataset for both the ML and DL models, which speaks to the framework's ability to estimate $PM_{2.5}$ reliably even as time interval changes, which is a relevant

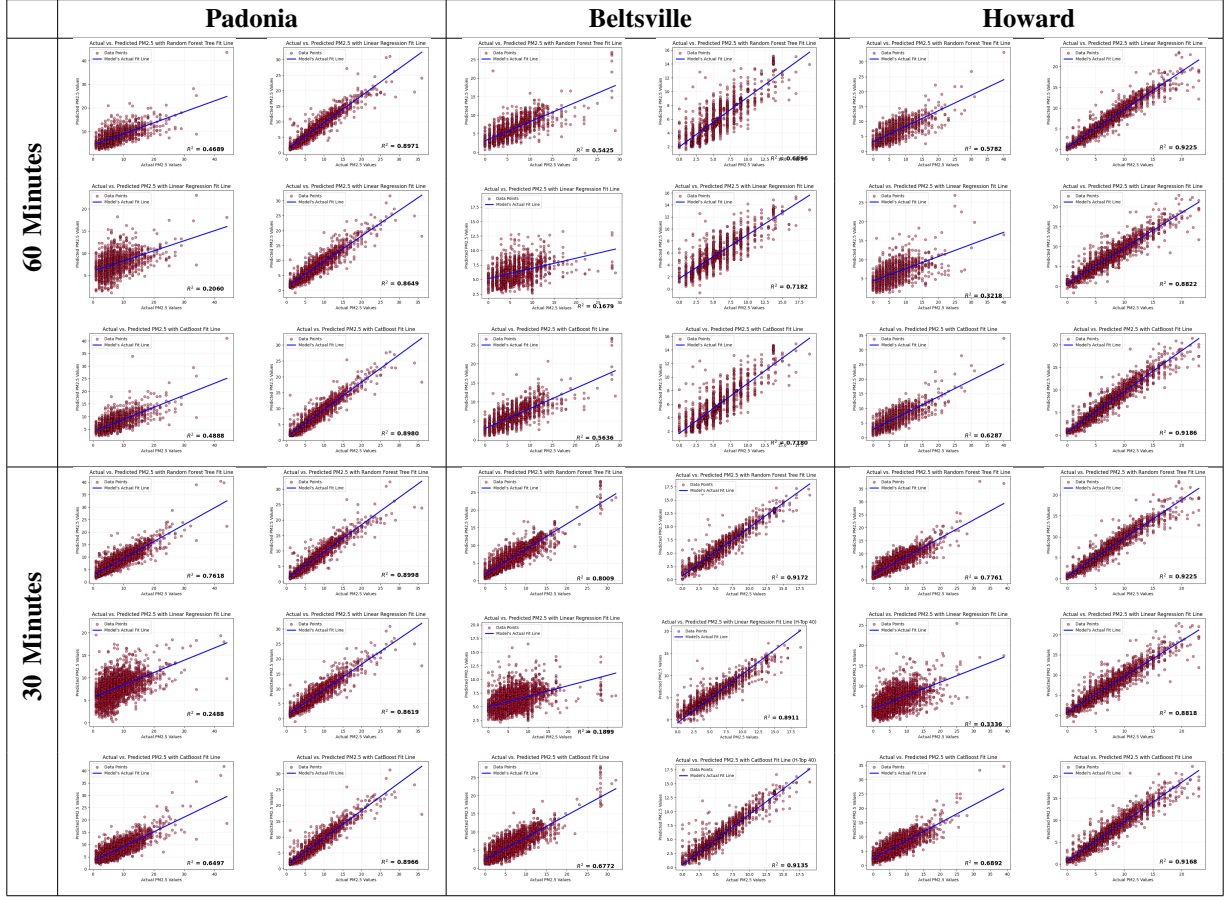


Fig. 2: Comparison of R^2 for ML models across three monitoring stations using hourly and half-hourly datasets.

consideration for regions where monitoring data may be sparse or irregular. The finer resolution preserved short-term changes in $PM_{2.5}$ that may have been smoothed out during hourly averaging, and this was especially noticeable for the DL models. At Howard, for example, SimpleRNN improved from an R^2 of 0.528 at the 60-minute interval to 0.836 at the 30-minute interval. Because models such as LSTM and GRU learn from sequences of previous observations, more frequent measurements allowed them to capture changes in $PM_{2.5}$ more precisely within the 10-step window. Although the DL models improved at the 30-minute interval, the feature-engineered ML models performed better overall, which has a direct bearing on the framework's practical goal of estimating $PM_{2.5}$ where monitoring infrastructure is limited. The highest DL result was an R^2 of 0.884 for Bi-GRU at Beltsville, while some ML models achieved higher scores, including Lasso Regression at Beltsville with an R^2 of 0.963. This difference is likely due to the DL models using only previous $PM_{2.5}$ values, while the ML models incorporated meteorological variables, AOD, lag features, and rolling statistics along with past

$PM_{2.5}$ models. Since the dataset covered only three sites from 2019 to 2022, it may also not have been large or complex enough for the recurrent models to show their full advantage. Taken together, these results suggest that for $PM_{2.5}$ prediction at this scale, directly engineering temporal and meteorological features is the more effective path toward the framework's original aim, which is producing accurate, transferable $PM_{2.5}$ estimates in locations lacking dense ground-based monitoring.

VI. LIMITATIONS

Despite the promising results, this study has several limitations. First, the dataset contained missing values and temporal gaps, particularly during nighttime, which may have reduced the models' ability to capture continuous temporal patterns despite extensive preprocessing. Second, the models were developed using data from only three monitoring stations in Maryland, limiting their generalizability to other geographic regions. Finally, the prediction framework relied on a limited set of features (AOD, temperature, relative humidity, wind speed, and lagged $PM_{2.5}$) and four years of data, which may not

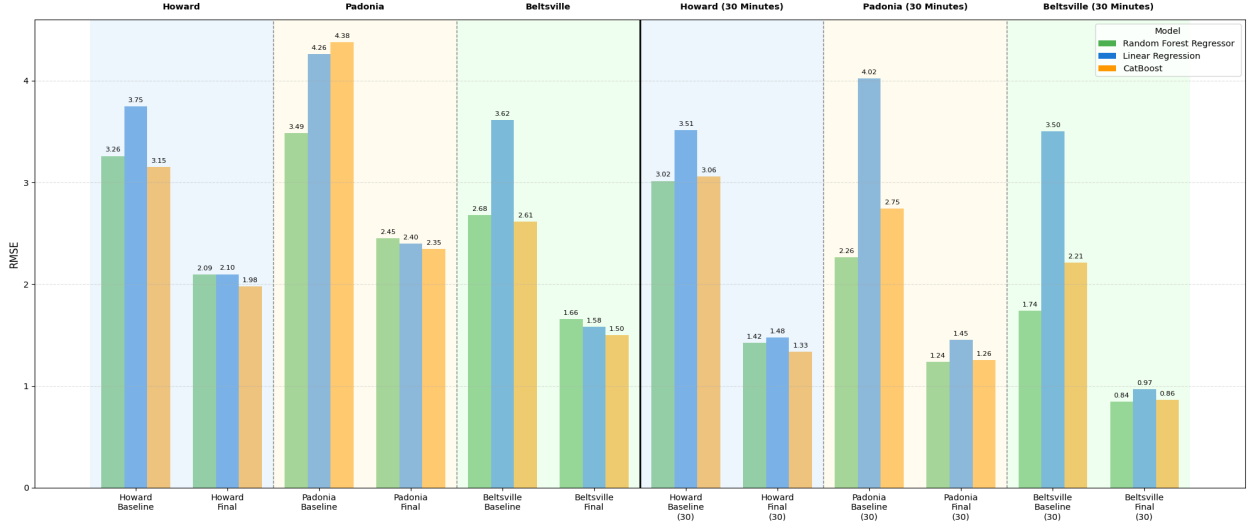


Fig. 3: RMSE comparison of ML models under different temporal resolutions and feature engineering phases.

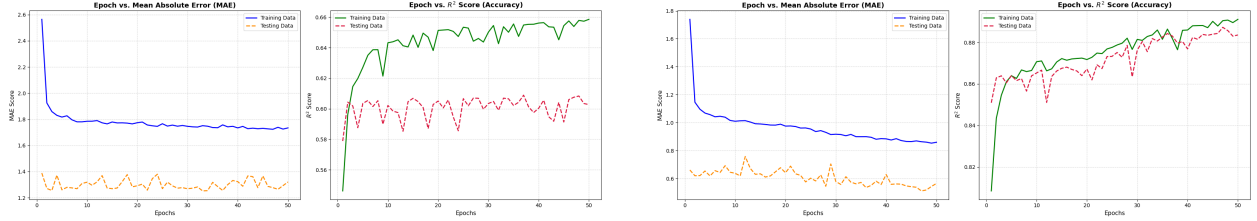


Fig. 4: Comparison of Bi-GRU results for the Beltsville 1-hour dataset (left) and 30-minute dataset (right).

fully represent the variability of long-term atmospheric conditions.

VII. FUTURE WORK

Future work will focus on improving data completeness by incorporating longer-term observations and reducing temporal gaps, particularly during nighttime. Expanding the study period beyond 2022 and including additional monitoring stations from diverse geographic regions will improve the temporal and spatial generalizability of the models. Future studies will also investigate additional variables, including wind direction, air pressure, planetary boundary layer depth, solar radiation, land use, traffic, emissions, and other air pollutants. Furthermore, advanced DL architectures, such as Transformer-based models, will be explored to further improve $PM_{2.5}$ prediction accuracy.

VIII. CONCLUSION

This paper explored the use of artificial intelligence in predicting $PM_{2.5}$ concentrations. The primary focus was training and testing machine learning and deep learning models based on weather observations and satellite-derived Aerosol Optical Depth (AOD) data collected in the District of Columbia, Maryland, and Virginia (DMV)

region. The data was split into two datasets with observation intervals of 60 minutes and 30 minutes, which were then synced, integrated, and processed. These datasets were used to test several machine learning models and five deep learning models: Simple RNN, LSTM, GRU, Bi-LSTM, and Bi-GRU.

The results showed that deep learning models outperformed the machine learning models that were evaluated. Deep learning models effectively captured the patterns in $PM_{2.5}$ data over time. Among the deep learning models, Bi-GRU had the best performance on the 60-minute dataset, while LSTM gave the highest predictive performance on the 30-minute dataset. The better results of the 30-minute dataset suggest that a higher temporal resolution can improve $PM_{2.5}$ predictions. Overall, these findings indicate that combining meteorological variables, satellite-derived AOD data, and deep learning offers a robust approach to predicting $PM_{2.5}$ concentrations. This can help with more accurate air quality monitoring and better environmental and public health decisions.

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